

REQUIREMENTS DOCUMENT

This slide is to describe System Documentation (SD) Template and also Software Requirements Specification (SRS) for Plan-driven and Agile based system development

System Documentation SD Template

Document Content

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- You are given a SD Template which contains
Revision page
Table of contents which have section 1-8. The description for the sections:
 - Section 1 – 2 : Introduction and Specific Requirements
This section represents the SRS for **A2**
 - Section 3-7: System Architectural Design until Requirements Matrix
This section represents the SDD for **A3**
 - Section 8: Test Cases
This section represents the STD for **A4**

Process to complete the System Documentation Template

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- When you have identified and confirm the features you want to develop from A1, you can start doing A2.
- When you completed A2, submit A1 (correction) together with A2 (Section 1-2)
- When you completed A3, submit A2 (1-2 update with correction) together with A3 (Section 3- 7)
- When you completed A4, submit together with A2 (section 1-2), A3 (Section 3- 7 update with correction) together with A4 (Section 8)

Software Requirements Specification (SRS)

**SRS sections which are the same in
both SRS Plan Driven and SRS Agile**

Revision Page (the same in both SRS Plan Driven and SRS Agile)

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Revision Page

a. Overview

Describe the content of the current version of SD (see below – the note in red).

b. Target Audience

State the targeted audience for the SD.

c. Project Team Members

List the team members in a table by stating their roles and the status for each assigned task e.g. by sections for this SD version (complete, partially complete, incomplete). If the assigned tasks are not done and have been assigned to other team members, state accordingly.

Member Name	Role	Task	Status

As a team you have to show how the task are distributed and explain truthfully which team member who did not participate in completing the document



d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Team leader 1 (Full Name)	Completed Chapter X, Section...	dd/mm/yyyy
...			

Since you completed the system documentation incrementally you have to show the versions of the document



Section 1

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1. Introduction

The following sections of the System Documentation (SD) should provide the details of the entire document comprising System Requirements Specification (SRS), System Design Document (SDD), and System Testing Document (STD). Remove the notes in red texts when submitting including these notes.

1.1 Purpose

This subsection should:

- a) *Delineate the purpose of the SD;*
- b) *Specify the intended audience for the SD.*

[The SD describes...]

1.2 Scope

This subsection should:

- a) *Identify the software product(s) to be produced by name (e.g., Online Learning System, Emotion Chatbot Analyzer, etc.);*
- b) *Explain what the software product(s) will, and, if necessary, will not do;*
- c) *Describe the application of the software being specified, including relevant benefits, objectives, and goals;*

[The software product is...]

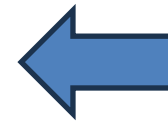
1.3 Definitions, Acronyms and Abbreviation

This subsection should provide the definitions of all terms, acronyms, and abbreviations used in the SD.

[Definitions of all terms, acronyms and abbreviations used are to be defined here.]

Follow the instruction in red to complete each Subsections. Make sure complete all the sections clear and precise with a short sentence

Make sure you indentify a suitable acronyms from our domain area
For example what is acronym DASS represent for the domain of mental health



Section 2.4 Performance and Other Requirements

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2.4 Performance and Other Requirements

Non-functional requirements are the requirements that describe how the system should behave or operate, rather than what it should do. They address the characteristics or qualities of the system, such as its usability, reliability, performance, security, maintainability, and compatibility. Non-functional requirements are important to write in the SD because they help ensure that the system meets the stakeholders' needs and expectations. The ISO/IEC/IEEE 29148 standard suggests that non-functional requirements should be specified under three main categories: Software System Attributes, Performance and Other Requirements.

Example of Software System Attributes are as follows:

Software system attributes define the overall qualities or characteristics of the software. These attributes are the foundation on which the software is built, and they include the following:

- *Usability: The ease with which users can interact with the system.*
- *Reliability: The ability of the system to perform its functions correctly and consistently.*
- *Maintainability: The ease with which the system can be modified, repaired, and enhanced.*
- *Portability: The ability of the system to run on different hardware and software platforms.*
- *Compatibility: The ability of the system to work with other systems and components.*

Example of Performance is as follows:

Performance requirements define the system's capability to respond to user requests and handle data in a timely manner.

These requirements include the following:

- *Response Time: The time it takes for the system to respond to a user request.*
- *Throughput: The number of requests the system can handle in a given period of time.*
- *Capacity: The maximum number of users or amount of data that the system can handle.*
- *Availability: The percentage of time the system is operational and accessible to users.*

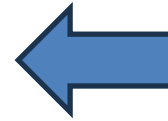
Example of Other Requirements is as follows:

Other requirements refer to any non-functional requirements that do not fit under the categories of software system attributes or performance. These requirements include the following:

- *Security: The protection of the system and its data from unauthorized access and malicious attacks.*
- *Safety: The ability of the system to operate safely without causing harm to users or the environment.*
- *Legal and Regulatory: Compliance with laws, regulations, and standards.*
- *Environmental: The impact of the system on the environment.*



This section is for you to describe your **PRODUCT** non functional requirements (NFR) Make sure you identify the **MOST SUITABLE** NFR and do not choose all the NFR because in real implementation NFR is implemented for the whole system and its costly to develop



This section is for you to describe your **ORGANIZATIONAL** and/ or **EXTERNAL** non functional requirements (NFR)

2.5 Design Constraints

Explain any constraints imposed by the organization where the software product will be used such as the system must adhere to certain organizational standard and other external non-functional requirements.

Example is as follows:

Environmental constraints: These constraints relate to the physical environment in which the system will operate, such as temperature, humidity, and lighting conditions. An example of an environmental constraint is: "The system must be designed to operate in temperatures ranging from 0°C to 40°C."

Hardware constraints: These constraints relate to the specific hardware components that will be used in the system, such as processors, memory, and storage. An example of a hardware constraint is: "The system must be designed to run on a minimum of 8GB of RAM."

Security constraints: These constraints relate to the security requirements for the system, such as data encryption, authentication, and access control. An example of a security constraint is: "The system must be designed to encrypt all sensitive user data using AES-256 encryption."

Compatibility constraints: These constraints relate to the compatibility requirements for the system, such as compatibility with specific software or hardware components. An example of a compatibility constraint is: "The system must be designed to be compatible with Windows 10 operating system."

Software Requirements Specification (SRS) Plant-Driven

SRS for Plan-Driven based system development

Section 2. Specific Requirements

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2. Specific Requirements

*This section of SD serves as the SRS that should contain all the software requirements to a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. Throughout this section, every stated requirement should be externally perceivable by users, operators, or other external systems. These requirements should **include at a minimum a description of every input (stimulus) into the system, every output (response) from the system, and all functions performed by the system in response to an input or in support of an output.***

2.1 User characteristics

This should specify the following:

In an SRS, the user characteristics section describes the intended users of the system and their characteristics, such as their technical expertise, knowledge, and physical abilities. This section is important because it provides the development team with a clear understanding of whom the system is designed for, and helps to ensure that the system is developed to meet the needs of its intended users.

Example. Describe how the system will interact with its users.

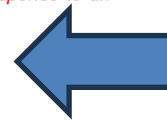
The "Emotion Chatbot Analyzer" software will be used by two main types of users: UTM students and UTM counselors.

2.1.1 UTM Students

- *The UTM students who will use the software are expected to have basic computer skills, including familiarity with web-based applications.*
- *They may have varying levels of emotional intelligence and may experience a wide range of emotional states, from mild stress to severe anxiety or depression.*
- *Some students may be reluctant to seek counseling or may not have access to professional counseling services, making the chatbot a valuable resource for emotional support and guidance.*

2.1.1 UTM Counsellors

- *The UTM counsellors who will use the software are expected to have professional training and experience in counselling and psychology.*
- *They may have varying levels of technical expertise, but will receive training on how to use the software.*
- *They will use the software to access student emotion records and provide counselling services to students as needed.*
- *Both types of users will be required to provide personal information such as name and email address in order to register for the software. The software will be designed with privacy and security measures in place to protect user information.*



Follow the description
given to describe user characteristics

Sub-section 2.2 System Features

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2.2 System Features

Specify the product perspective.

Example.

The Emotion Chatbot Analyzer is a software system that operates on mobile devices, including both Android and iOS operating systems. The system provides a means for UTM students to better manage their emotional health and at the same time simplifies the process of seeking emotional support from UTM counselors. The system also implemented a real-time cloud-based database for better user experiences.

Specify the product module and function..

Include use case diagram here – see an example of Emotion Chatbot Analyzer below

Then, provide the description of each module and function using a table.

Example.

The system features is illustrated in Figure X.X below. The detail description of each module and functions is tabulated in Table X.X.

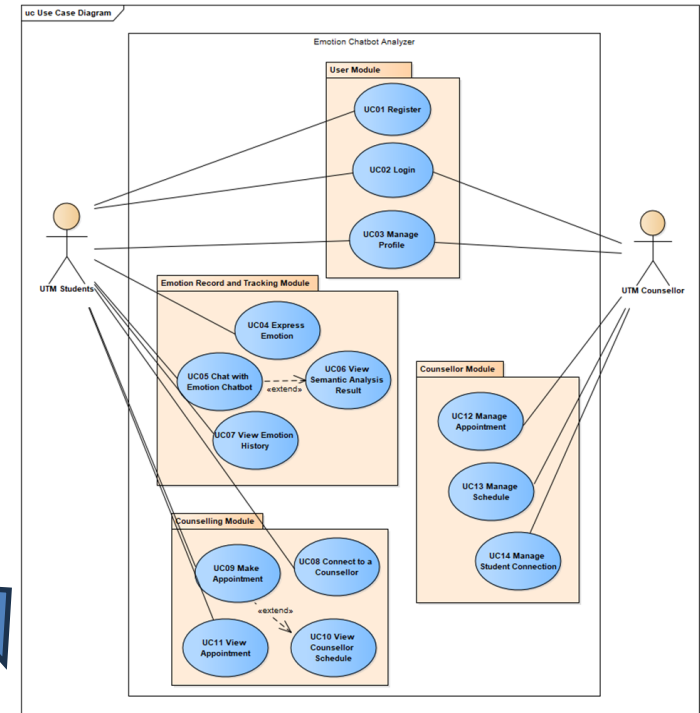


Figure X.X: Use Case Diagram for <Name of the System>

Table X.X: Description of Module and Functions for <Name of the System>

Module	Function	Description
User Module	UC01 – Register	This use case allows students to sign up as a user for the mobile application.

Grouping use case should reflect the modularity of the system for easier implementation and maintenance. To determine which use case belongs to which module (though this is not an absolute method):

- If one use case is an extension of another use case
- If use cases share a common goal or contribute to a specific functionality, group them in one module.
- If each module has a specific purpose and serves different actors

Sub-section 2.2 System Features

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[Include domain model i.e. **class diagram** without the operation part – only attributes without details on visibility and type, explain each class and its attributes including the relationships among the classes, see the example.]

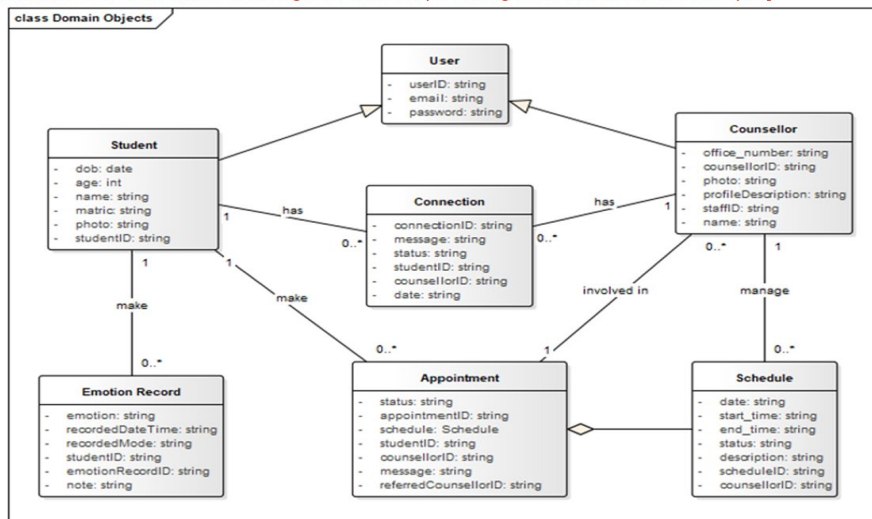


Figure X.X: Domain Model for <Name of the System>

To create a Domain Model for your system you can do the following:

- Review system descriptions, use cases, or other documentation and extract nouns as potential candidates for classes.

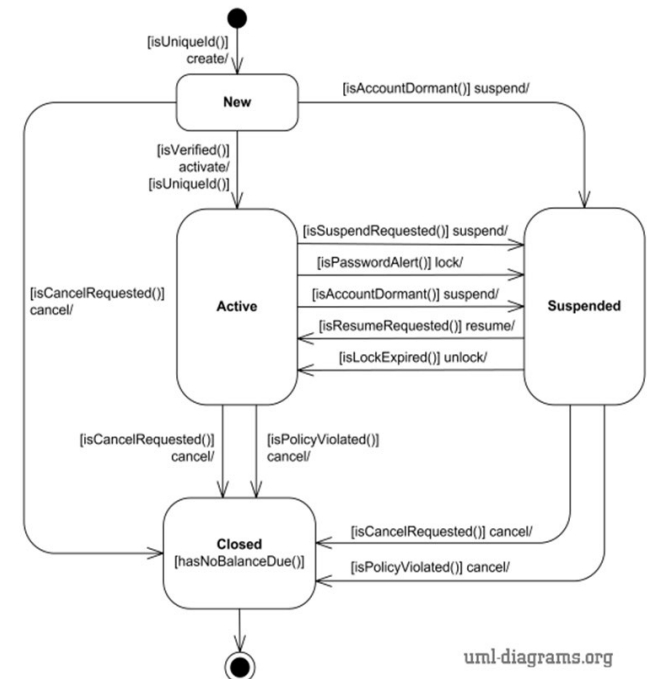


Figure X.X: State Diagram for <Name of the Class>

To create state diagram for your system you can identify a class which is significant within the domain. It should represent an entity or process that the system needs to manage or track.

Sub-section 2.3 Use Case Details

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2.3 Use Case Details

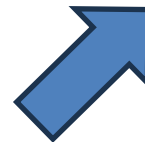
2.3.1 UC001: Use Case <Name of Use Case 1>

[Provide code for each use case such as UC001 and so on... Include Use Case Description for each use case as in the example below. If there are any alternative flows, add the rows accordingly. Otherwise, just remove the rows. Ensure each use case has its own unique ID and the name of the heading above corresponds to the name of use case in the use case diagram (refer to Figure X.X). Different scenarios of a use case require separate use case descriptions.]

Table X.X: Use Case Description for <Name of Use Case>

Use case: <Name of Use Case>
ID: UCxxx
Actors:
Preconditions:
Flow of events:
1.
2.
...
Postconditions:
Alternative flow n:
Postconditions:
Exception flow (if any):

[Include sequence diagram and activity diagram for each respective user case. See example below. May consider including different scenarios in different diagrams, if necessary, to avoid clutter.]



If you have 14 use cases as shown in the Emotion Chatbot Analyzer System, you must have 14 use case descriptions (UCD) with suitable sequence diagram and activity diagram for each UCD,

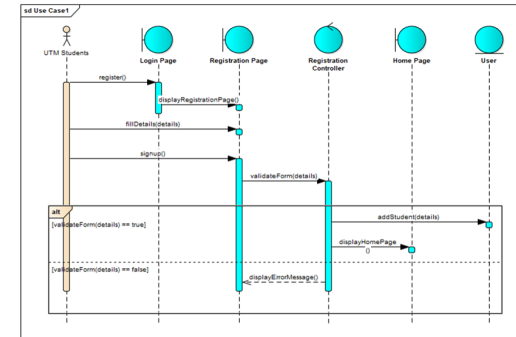


Figure X.X: Sequence Diagram for <Name of Use Case/Scenario if more than one scenario>

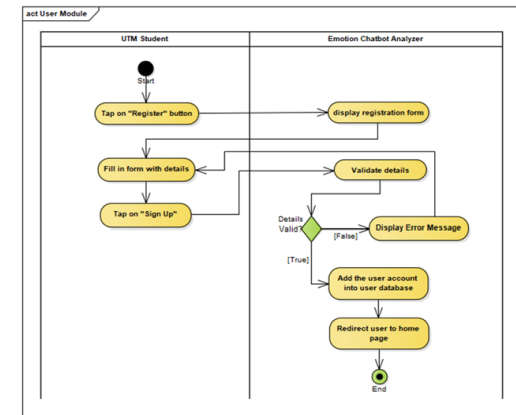


Figure X.X: Activity Diagram for <Name of Use Case/Scenario if more than one scenario>

Software Requirements Specification (SRS) Agile

SRS for Agile based system development

Sub-Section 2.1.1 Persona

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2.1 Persona

Describes the intended users of the system and their characteristics, such as their technical expertise, knowledge, and physical abilities. This section is important because it provides the development team with a clear understanding of whom the system is designed for, and helps to ensure that the system is developed to meet the needs of its intended users.

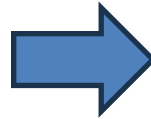
2.1.1 Persona 1 (User type 1/Actor 1)

User Need

- A user need can follow a template like: <user type> needs a way to <achieve a goal>

User Stories

- User stories are short descriptions of a feature told from the perspective of the user who desires the new capability.
- A user story generally follows a template like: As a <user type>, I want <feature/interface/functionality> so that <reason>
- Sometimes, a user story may cover a large amount of functionality. In these cases, these stories may be referred to as 'epics' and may need to be broken out into individual, smaller user stories.



Example of persona based on the template (following the use case diagram for Emotion Chatbot Analyzer System)

Describe persona for every user/actor for your system

2.1.1 Persona 1: UTM Students

- **User Need:** UTM students need a way to manage their emotional health, access counselors, and track their emotional well-being.
- **User Stories:**
 - As a UTM student, I want to register and log into the Emotion Chatbot Analyzer so that I can access its features.
 - As a UTM student, I want to express and track my emotions so that I can better understand and manage my emotional health.
 - As a UTM student, I want to schedule counseling appointments so that I can receive professional guidance.

Sub-section 2.2 System Features

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2.2 System Features

Specify the product perspective.

Example.

The Emotion Chatbot Analyzer is a software system that operates on mobile devices, including both Android and iOS operating systems. The system provides a means for UTM students to better manage their emotional health and at the same time simplifies the process of seeking emotional support from UTM counselors. The system also implemented a real-time cloud-based database for better user experiences.

Specify the product module and function..

Include use case diagram here – see an example of Emotion Chatbot Analyzer below

Then, provide the description of each module and function using a table.

Example.

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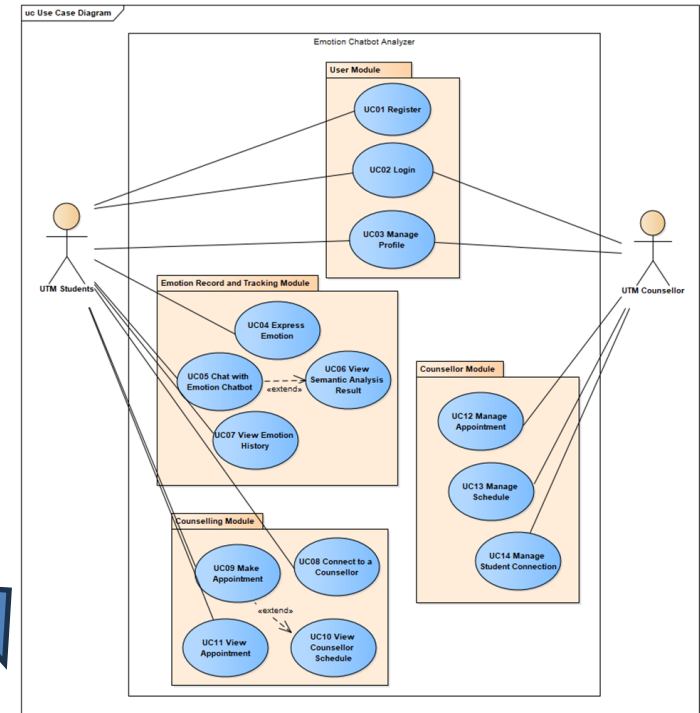


Figure X.X: Use Case Diagram for <Name of the System>

Table X.X: Description of Module and Functions for <Name of the System>

Module	Function	Description
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Grouping use case should reflect the modularity of the system for easier implementation and maintenance. To determine which use case belongs to which module (though this is not an absolute method):

- If one use case is an extension of another use case
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Sub-section 2.2 System Features

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[Include domain model i.e. **class diagram** without the operation part – only attributes without details on visibility and type, explain each class and its attributes including the relationships among the classes, see the example.]

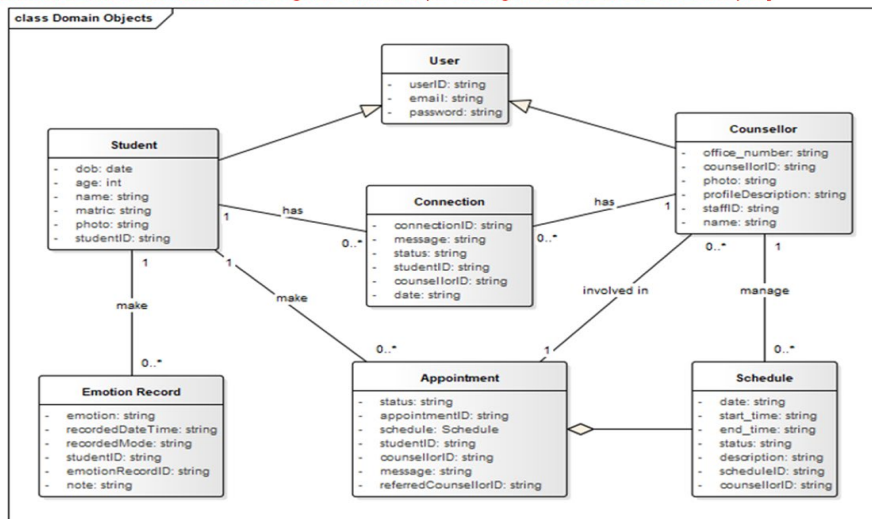


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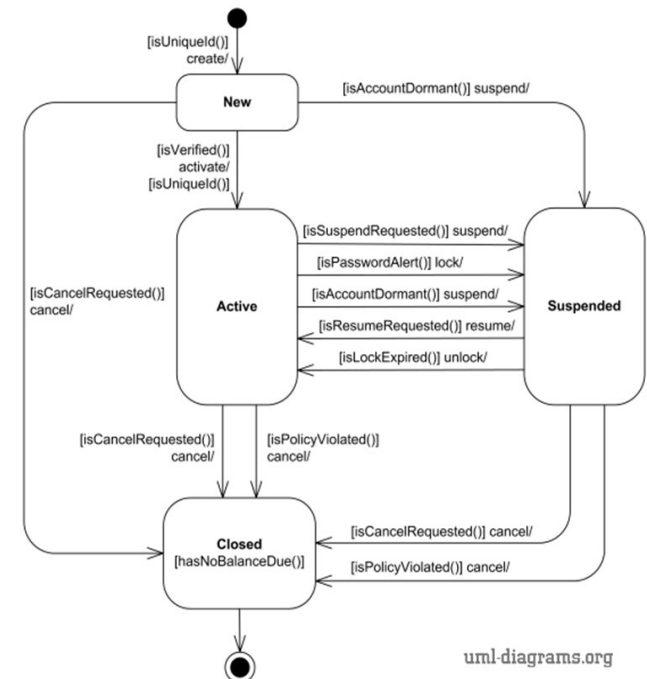


Figure X.X: State Diagram for <Name of the Class>

To create state diagram for your system you can identify a class which is significant within the domain. It should represent an entity or process that the system needs to manage or track.

Sub-section 2.3 System Features

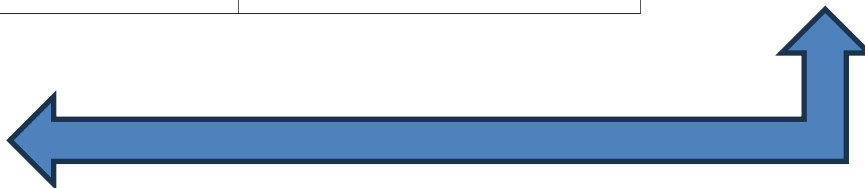
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2.3 Launch Phase

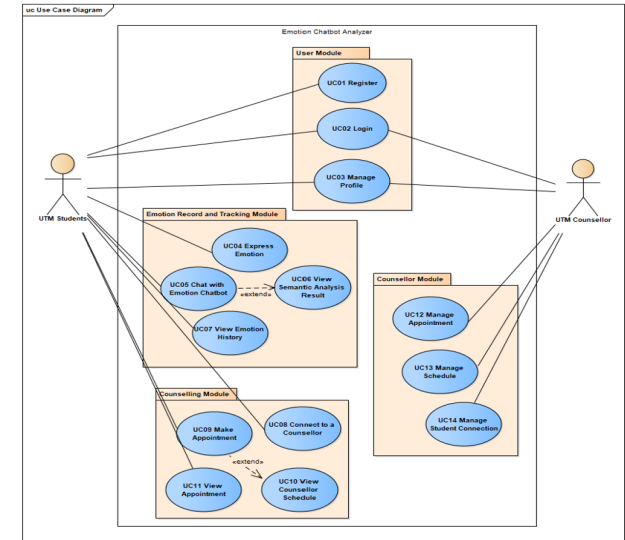
Product Backlog [describe more on product backlog];

Sprint	Team members assigned
Sprint #1 <i>Which user story assigned to this Sprint</i> [you can also specify based on which module you want to develop for each sprint – based on your use case diagram]	
Sprint #2 <i>Which user story assigned to this Sprint</i>	
Sprint #n <i>Which user story assigned to this Sprint</i>	

State who are assigned to do the sprint



You can list which user story/ module that you want to develop 1st 2st and so forth in the product backlog



2.1.1 Persona 1: UTM Students

- **User Need:** UTM students need a way to manage their emotional health, access counselors, and track their emotional well-being.
- **User Stories:**
 - As a UTM student, I want to register and log into the Emotion Chatbot Analyzer so that I can access its features.
 - As a UTM student, I want to express and track my emotions so that I can better understand and manage my emotional health.
 - As a UTM student, I want to schedule counseling appointments so that I can receive professional guidance.

Section 2.4 User Story Details

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2.4 User Story Details

[Provide ID for each user story such as US001 and so on...]

Include User Story Description for each use case as in the example below. If there are any alternative flows, add the rows accordingly. Otherwise, just remove the rows. Ensure each User Story has its own unique ID and the name of the heading corresponds to the name of use case in the use case diagram (refer to Figure 2.1). Different scenarios of a user story require separate descriptions.]

2.4.1 US001: User Story <User Story 1>

Table 2.1: User Story Description for <Name of Use Case>

User story: <Name of Use Case>
ID: USxxx
User Story Description
As a customer of the bank I want to be able to log into the system So that I can use bank products
Flow of events:
1.
2.
3.
...
Alternative flow n:
Acceptance Criteria
Postcondition
precondition
other conditions
Exception flow:
(if any in the event of error)

2.4.1 US001: Register

- **User Story:** As a UTM student, I want to register for the Emotion Chatbot Analyzer so that I can access its features.
- **Flow of Events:**
 1. The student accesses the registration page.
 2. The student inputs their details (e.g., name, email, and student ID).
 3. The system verifies the input and creates an account.
- **Alternative Flow:**
 - *Invalid details:* The system prompts the user to re-enter information.
- **Postcondition:** The student account is created successfully.
- **Precondition:** The student has not already registered.

Sample User Story based on the template

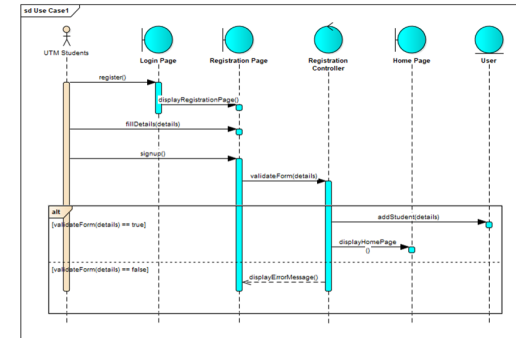


Figure X.X: Sequence Diagram for <Name of Use Case/Scenario if more than one scenario>

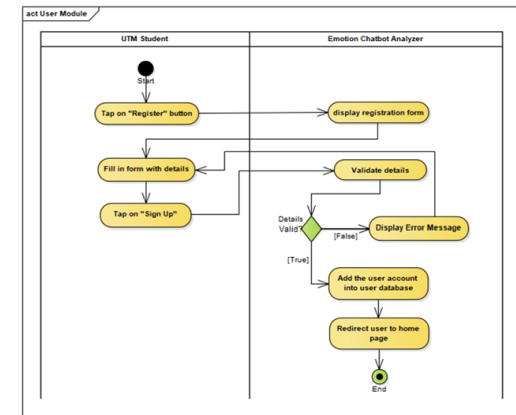


Figure X.X: Activity Diagram for <Name of Use Case/Scenario if more than one scenario>

If you have 14 user story/use cases as shown in the Emotion Chatbot Analyzer System, you must have 14 User Story Details with suitable sequence diagram and activity diagram for each UCD,